Still-Image Smoke Detection Using Colour and Texture Features with Supervised Machine Learning for Early Fire Alert Systems

PROJECT REPORT

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by

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ABSTRACT

This project presents a Python-based prototype for detecting smoke in individual still images. Inspired by a background-subtraction and Fuzzy C-Means (FCM) approach, it replaces frame differencing for still images with candidate-region generation based on HSV colour thresholds, local edge density, morphological operations and connected components. Seven colour, shape and texture features are extracted from each candidate. A trained Random Forest classifier is used. A separate video mode uses frame differencing, and the system can raise a terminal alarm and send a push notification through ntfy.

The reported evaluation covered 1,332 D-Fire images and 1,400 ground-truth smoke boxes. At an intersection-over-union (IoU) threshold of 0.3, the system recorded 472 true positives, 921 false positives and 928 false negatives, giving 33.88% precision, 33.71% recall and an F1-score of 33.80%. The results demonstrate an end-to-end prototype, while the number of false detections and missed regions shows that localization and classification require improvement. The prototype is not suitable as a standalone life-safety alarm.

Keywords: smoke detection, still images, HSV colour thresholding, edge density, Fuzzy C-Means, Random Forest, D-Fire dataset, ntfy notification.

1. Introduction

1.1 Background and Motivation

Smoke may become visible before flames, so early detection can provide additional time for people to respond. Conventional sensor-based detectors are effective in many enclosed spaces but must be placed near the source and may be expensive to deploy across large or open areas. Camera-based detection can use existing image sources, including surveillance cameras, drone images and publicly available photographs.

Smoke is visually variable. It can appear white, grey or dark depending on lighting and may resemble clouds, haze, fog or pale objects. Colour-only rules can therefore produce false alarms or miss smoke in difficult conditions. The project investigates whether combining colour, edge, shape and texture features with machine learning can improve the classification of candidate smoke regions in still images.

1.2 Related Work

Vijayalakshmi and Muruganand [1] used frame differencing to identify moving regions in video and Fuzzy C-Means (FCM) clustering to distinguish smoke from other motion. Their method relies on a known background and is not directly applicable to a single photograph. The D-Fire dataset [2] provides annotated smoke and fire images for supervised learning and quantitative evaluation. This project combines heuristic candidate-region extraction, handcrafted features, Random Forest classification.

1.3 Project Overview

For still images, the system generates candidate regions using HSV colour and soft-edge cues, extracts seven features and classifies each region. When any region is labelled as smoke, the program produces a terminal alert and attempts to send a high-priority ntfy notification. A separate video mode implements frame differencing and periodically applies the image detector.

2. Problem Statement, Objectives and Scope

2.1 Problem Statement

To develop and evaluate a still-image smoke-detection pipeline that identifies candidate regions without requiring a background frame, describes them using colour, shape and texture features, classifies them using FCM or a trained Random Forest, evaluates detections against ground-truth boxes, and issues an alert when smoke is detected.

2.2 Objectives

1. Generate candidate smoke regions from an individual image without a background frame.
2. Extract seven colour, edge, shape and texture features for each candidate region.
3. Use a Random Forest model when available and FCM as a fallback classifier.
4. Evaluate detections using IoU, precision, recall and F1-score.
5. Generate a terminal alarm and attempt a phone push notification when smoke is detected.

2.3 Scope

In scope: still-image smoke detection using a subset of D-Fire; candidate-region extraction and feature-based classification; quantitative evaluation against smoke bounding boxes; a prototype ntfy alert; and a separate frame-differencing video mode.

Outside scope: detecting flames as a separate class, deep-learning detection models, real-time video classification, and connecting the prototype to residents or emergency services.

3. Requirement Analysis

3.1 Functional Requirements

ID	Requirement
R1	Accept a still image from a file path.
R2	Generate candidate regions from colour and edge cues without a background frame.
R3	Calculate seven features for each candidate region.
R4	Use the saved Random Forest model if available; otherwise use FCM.
R5	Save masks, detections and visual output for each image.
R6	Raise a terminal alarm and send an ntfy push-notification request when smoke is detected.
R7	Train the classifier from labelled data and evaluate detections against ground truth.
R8	Provide a separate video mode using frame differencing.

3.2 Non-functional Requirements

- Run on a CPU-only laptop using open-source libraries.
- Use fixed parameters and random seeds for reproducibility.
- Keep classification decisions interpretable through a small set of handcrafted features.

3.3 Hardware and Software Requirements

Item	Specification reported in the paper
Computer	Windows laptop; processor and RAM not stated in the evaluation output.
Phone	Android phone with ntfy app and internet connection.
Language / environment	Python; Windows PowerShell used for evaluation..
Libraries	OpenCV, NumPy, scikit-learn and joblib.
Notification service	ntfy.sh.

3.4 Data Requirements

The D-Fire dataset contains more than 21,000 images with YOLO-format labels. Each label records class ID, normalized centre coordinates, width and height. In the reported experiment, class 0 was treated as smoke and class 1 as fire.

Data subset	Images / annotations	Use
Reduced evaluation working set	1,332 images	Evaluation run reported in the paper.
Training subset	400 images	Used to train the Random Forest.
Held-out images	932 images	.
Final reported evaluation	1,400 ground-truth smoke boxes	Detection metrics

The paper describes both a 400-image training / 932-image held-out setup and a final evaluation run on 1,332 images. The quantitative results in this report follow the paper's final reported evaluation table.

4. System Architecture and Design

4.1 Architecture

The system follows two paths. The detection path loads an image, generates candidate regions, extracts features, classifies candidates and triggers an alert if any region is labelled smoke. The training path uses D-Fire labels to label candidate regions, trains the Random Forest and saves the model. The evaluation process matches predicted boxes to ground-truth smoke boxes using IoU.

Stage	Operation	Output
1. Input	Load still image in BGR format.	Input image
2. Candidate generation	HSV colour mask, Canny edge map and local edge-density mask; morphological opening/closing and connected components.	Candidate regions
3. Feature extraction	Compute seven colour, edge, shape and texture features.	Feature vector per candidate
4. Classification	Random Forest if saved model exists; otherwise FCM with two clusters.	Smoke / non-smoke labels
5. Decision and alert	If any region is smoke, raise terminal alarm and send ntfy request.	Alert attempt and saved output
6. Evaluation	Match predicted boxes to ground-truth boxes at IoU ≥ 0.3.	Precision, recall and F1-score

4.2 Feature Design

Feature	Description	Purpose stated in the paper
Mean hue	Average hue in the region.	Colour information.
Mean saturation	Average saturation.	Smoke is often desaturated.
Mean value	Average brightness/value.	Colour and brightness cue.
Edge density	Fraction of Canny edge pixels in the region.	Smoke often has soft edges.
Solidity	Region area divided by convex-hull area.	Represents shape irregularity.
Extent	Region area divided by bounding-box area.	Additional shape cue.
Laplacian variance	Variance of the Laplacian within the region.	Estimates sharpness/texture; soft smoke is expected to have lower values.

4.3 Classification Design

FCM clusters normalized feature vectors into two groups ($c = 2$, fuzziness coefficient $m = 2$, termination tolerance $\varepsilon = 0.001$). The cluster with the lowest combined saturation, edge density and Laplacian variance is treated as smoke. The supervised alternative is a Random Forest with 100 trees, maximum depth 8, balanced class weights and random_state 42.

5. Implementation

5.1 Candidate-region Generation

For still images, the BGR input is converted to HSV and grayscale. The colour mask retains pixels with saturation $S < 50$ and brightness $90 < V < 235$. Canny edge detection uses thresholds 40 and 120; local edge density is calculated using a 15×15 window, and pixels below 0.12 are retained as possible soft-edge regions. The masks are combined and refined using a 9×9 elliptical morphological kernel with opening and closing. Connected components smaller than 600 pixels are discarded.

5.2 Feature Extraction

The seven features described are computed for every candidate. Laplacian variance is calculated from the grayscale image using a Laplacian operator with kernel size 3. This sharpness measure is an addition to the feature set described for this project.

5.3 Training and Classification

Candidate regions are labelled against ground-truth smoke boxes. A candidate is treated as smoke when its IoU with a ground-truth box is at least 0.05 or when it contains the centre of a ground-truth box. The centre rule addresses cases where candidate regions are larger than the annotation boxes. The Random Forest is trained on these labelled feature vectors and saved as models/smoke_classifier.joblib. At detection time, the saved model is loaded if present; otherwise FCM is used.

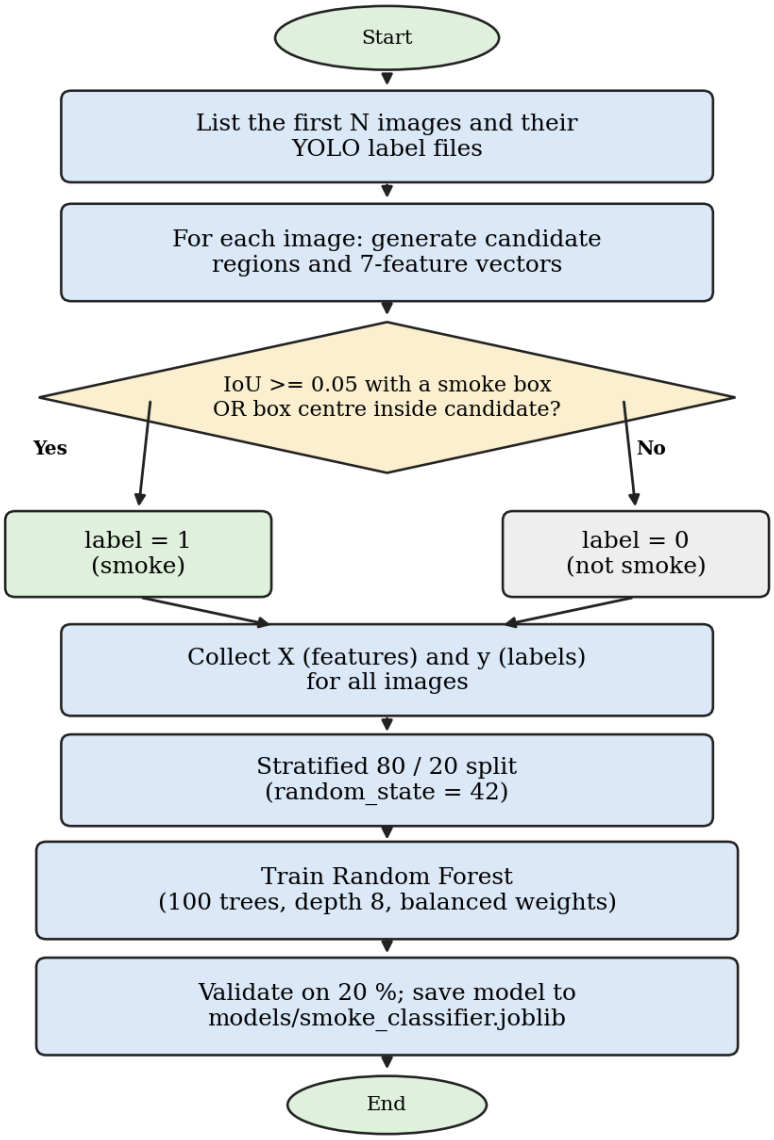


Fig 1(a): Flowchart of the training of the Random Forest classifier

5.4 Main Software Modules

Module	Purpose
main.py	Command-line entry point for image, evaluate, train and video operations.
background_subtraction.py	Candidate generation for still images and frame differencing for video.
features.py	Computes and normalizes the seven features.
fcm.py	Implements Fuzzy C-Means clustering.
train_classifier.py	Labels candidates using ground truth and trains/saves the Random Forest.
detect_image.py	Runs the still-image detection pipeline.
evaluate.py, utils.py, dataset.py	Ground-truth parsing, image/label pairing, IoU matching and metrics.
alert.py, phone_alert.py	Terminal beeps/banner and ntfy push-notification request.

5.5 Alert Mechanism

When smoke is detected, the program prints a terminal banner and attempts three Windows beeps where the winsound module is available. It sends an HTTP POST request to ntfy.sh using a configured topic, the title “FIRE / SMOKE EMERGENCY”, priority 5 and fire-related tags. A successful HTTP response means the service received the request; it does not prove that the phone played a sound. Phone sound and vibration depend on notification permissions, channel settings, volume and network connectivity.

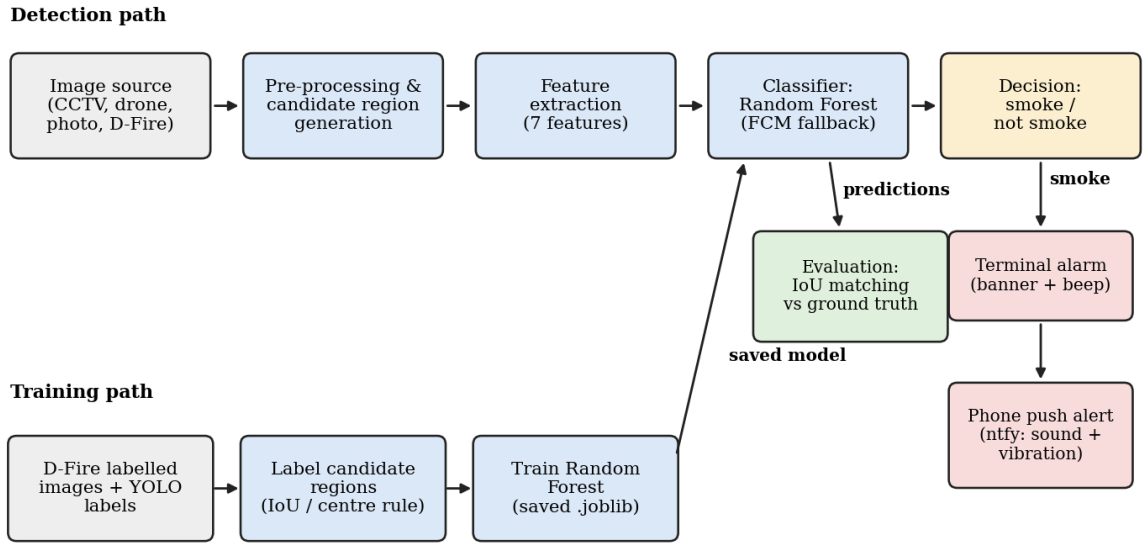


Fig 1(b): Block diagram of the smoke detection and alert system

5.6 Video Mode

The separate video mode converts frames to grayscale, uses the first frame to establish a background, and compares later frames with it using an absolute-difference threshold of 5. The background is updated with a learning rate of 0.02. Sample motion masks are saved periodically, and the image classifier is applied to every 15th frame by default. The paper notes that the motion mask is not directly combined with the still-image candidate mask.

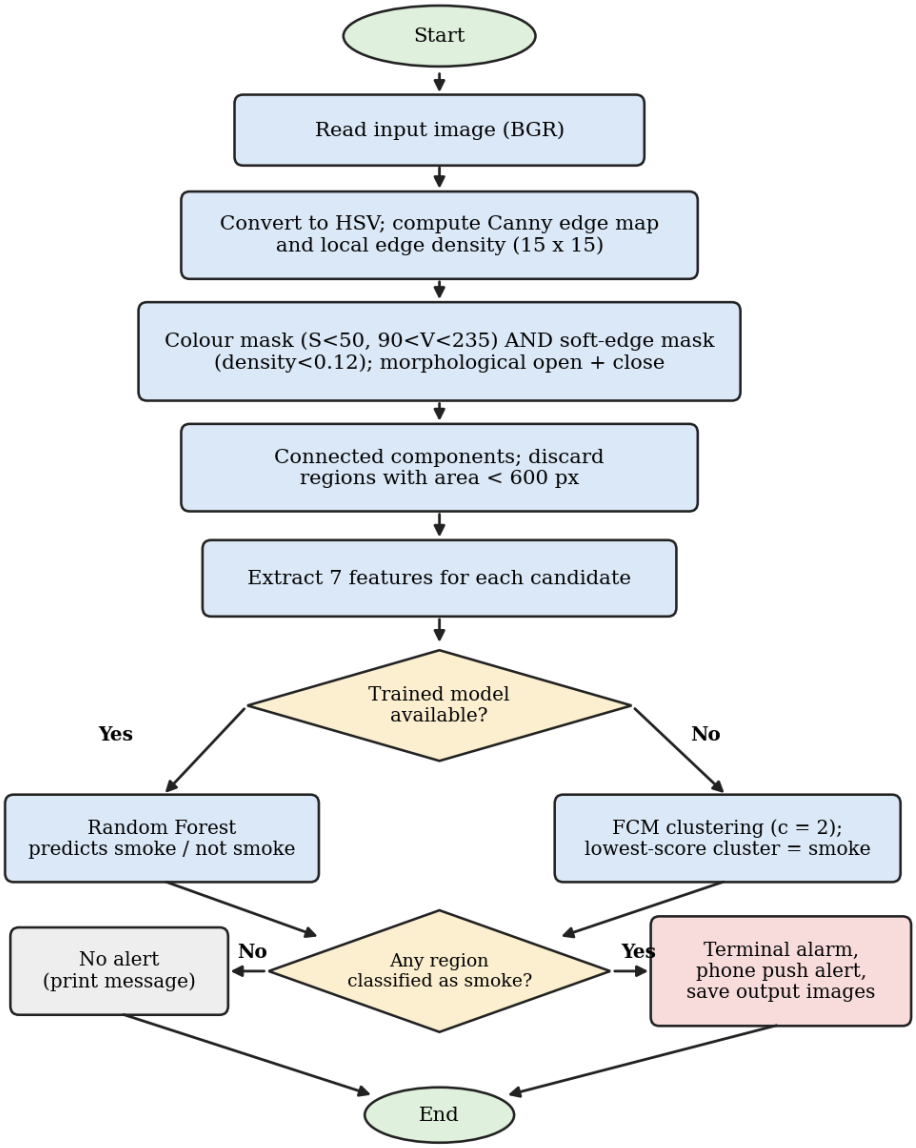


Fig 1(c): Flowchart of the smoke detection algorithm

6. Testing

6.1 Test Strategy and Metrics

Testing included synthetic images and video, visual checks of real images against labels, and quantitative evaluation. For the reported evaluation, a predicted box is a true positive when its IoU with an unmatched ground-truth box is at least 0.3. Each ground-truth box can be matched once. Unmatched predictions are false positives and unmatched ground-truth boxes are false negatives.

Precision = TP / (TP + FP); recall = TP / (TP + FN); F1-score is the harmonic mean of precision and recall.

6.2 Test Cases

Test case / input	Method	Observed result	Status
Synthetic grey blurred disc on green background	FCM	One candidate was flagged as smoke.	Pass
Hazy landscape with no smoke	FCM	Whole sky was flagged as smoke.	Fail — false positive
Small labelled wisp above a car park	FCM with tightened thresholds	15 candidates; 5 flagged, including a sky band and parked cars.	Partial
Cloudy sky without smoke	FCM	Cloud region was flagged as smoke.	Fail — false positive
Night fire with smoke plume	FCM	No candidate region; no alert.	Fail — false negative
Synthetic video, 90 frames	Frame differencing	Motion mask followed the growing blob.	Pass
D-Fire evaluation run	Random Forest pipeline	Precision 33.88%, recall 33.71%, F1 33.80% in the reported run.	Quantitative evaluation
Smoke-classified image	Terminal alarm and ntfy	Notification request is implemented; phone sound/vibration depend on device settings.	Delivery confirmation not established in paper

7. Results

7.1 Quantitative Results

The final reported evaluation used 1,332 images and 1,400 ground-truth smoke boxes. Results below are reported at IoU ≥ 0.3.

Metric	Reported value
Images evaluated	1,332
Ground-truth smoke boxes	1,400
True positives	472
False positives	921
False negatives	928
Precision	33.88%
Recall	33.71%
F1-score	33.80%

7.2 Qualitative Results and Error Analysis

The paper reports a correct detection of a smoke plume from a night-time house fire, showing that a smoke region can be detected in low light in at least one example. A large plume in an aerial view was not localized correctly: the output box appeared near the lower-left region rather than around the plume, producing a missed detection and a false positive. A rainy night street was correctly rejected, with wet-road reflections and street lights producing no candidate regions.

The high false-positive count indicates that many predicted boxes did not sufficiently overlap ground-truth smoke boxes at the selected IoU threshold. The false-negative count indicates that many annotated smoke boxes were missed. Fixed colour thresholds, candidate regions that are too large or poorly positioned, and visual similarities between smoke, clouds, haze and pale objects are identified as possible causes.



Fig. 2(a). Correct detection: night-time house fire.
The smoke plume is segmented in the mask and enclosed by a red box in the output. This is a true positive in low light.



Fig. 2(b). Failure case: large smoke plume from an aerial view.
The mask covers the plume, but the box in the output is on a small region near the bottom left instead of the plume. This is a missed detection plus a false positive, which shows the limits of the candidate stage and the classifier on large smoke regions.



Fig. 2(c). Correct rejection: rainy night street.
Wet-road reflections and street lights produce no candidate regions and no alert, so this is a true negative.

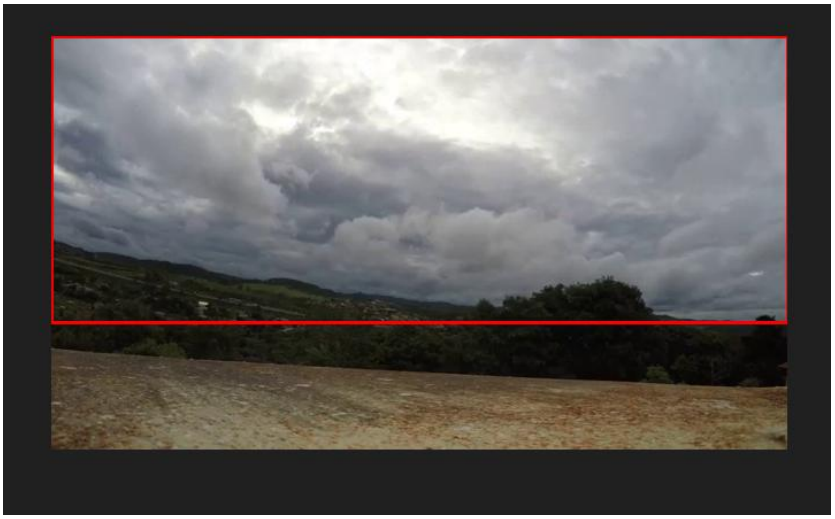


Fig 2(d): False positive: clouds flagged as smoke on an image with an empty label file



Fig 2(e): False positive: the sky of a clean landscape flagged as smoke

8. Limitations and Future Scope

8.1 Limitations

- Precision, recall and F1-score remain limited in the reported evaluation, with substantial false positives and missed smoke boxes.
- Fixed colour and edge thresholds can confuse smoke with clouds, haze and pale objects.
- Candidate regions can be too large or incorrectly positioned, causing poor bounding-box localization.
- The system detects smoke, not flames as a separate class.
- Phone alert sound and vibration are controlled by device notification settings and connectivity; a notification request does not guarantee audible delivery.
- The prototype is not a standalone life-safety alarm and is not connected to emergency services.

8.2 Future Scope

- Add or replace the candidate stage with a detector designed specifically for smoke and fire.
- Improve bounding-box localization and include difficult negative examples such as clouds and haze.
- Evaluate FCM and Random Forest separately on the same held-out test set.
- Integrate video motion masks more directly with smoke classification and test live camera feeds.
- Report reproducible dataset size, precision, recall, F1-score, average precision and inference time.
- Add alert-delivery status checks and clearer configuration of phone notification settings.

9. Conclusion

The project implements a still-image smoke-detection prototype using heuristic colour and edge thresholds to generate candidate regions, seven handcrafted features, and either a trained Random Forest classifier or an FCM fallback. It also includes a separate video mode based on frame differencing and an alert path using a terminal banner and ntfy push-notification request.

In the reported evaluation of 1,332 images at IoU ≥ 0.3, the system achieved 33.88% precision, 33.71% recall and an F1-score of 33.80%. These results demonstrate the end-to-end implementation but also show that false detections and missed regions remain substantial. Further work is needed to improve candidate generation and localization, evaluate methods under directly comparable conditions, and verify alert delivery. The prototype should not be used as a standalone life-safety alarm.

10. References

[1] S. R. Vijayalakshmi and S. Muruganand, “Smoke detection in video images using background subtraction method for early fire alarm system,” in Proceedings of the 2nd International Conference on Communication and Electronics Systems (ICCES), 2017, pp. 167–171.

[2] “D-Fire: An image dataset for fire and smoke detection,” GitHub repository. Available: https://github.com/gaia-solutions-on-demand/DFireDataset